

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

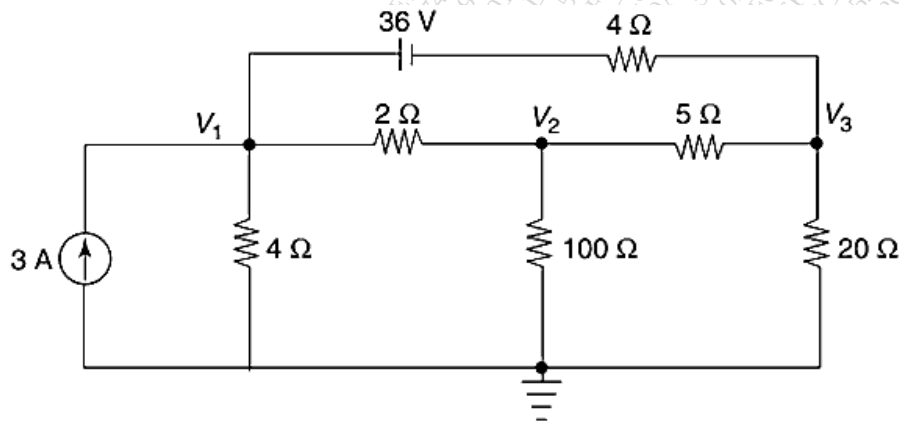
(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) For the network shown, find voltages V_1 and V_2 using Node Analysis

CO1

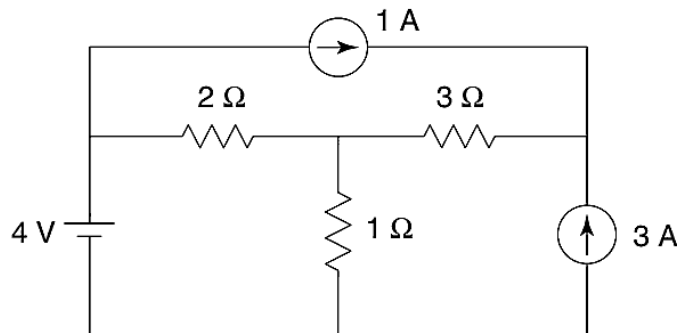
6



- B) State Superposition theorem and Find the current in the 1 Ω resistors

CO1

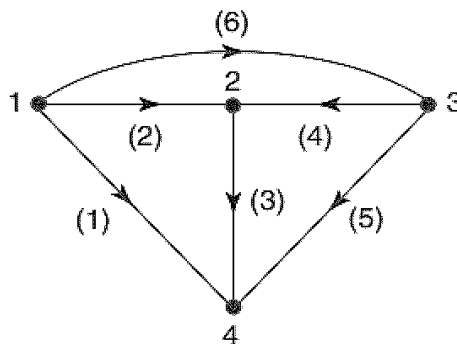
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- C) Define the terms : Branch and Node related to graph of a network and For the graph shown below write the (a) incidence matrix (b) Tie set matrix

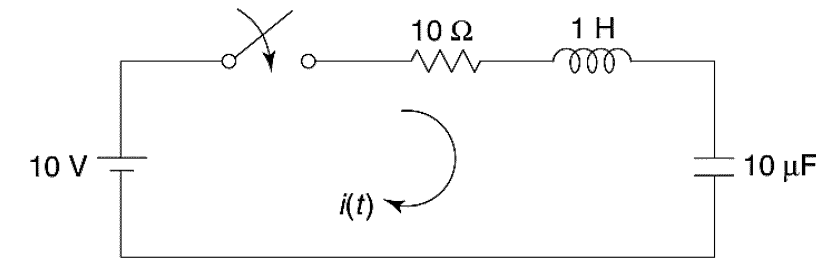
CO1

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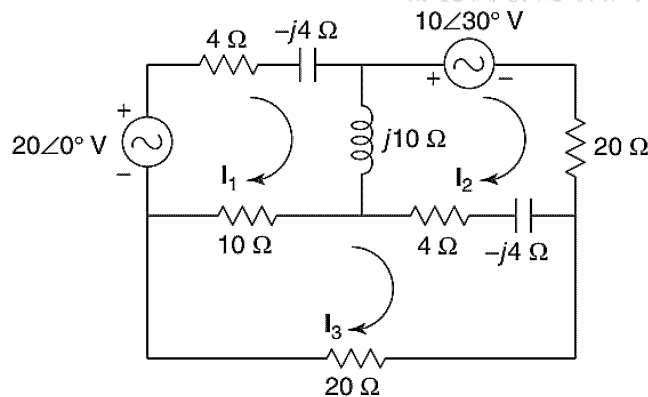


Q.2 Solve Any Two of the following.

- A) In the given network of Fig., the switch is closed . Assuming all Initial **CO1, CO3** **6**
conditions as zero, find i , di/dt and d^2i/dt^2 at $t = 0^+$.



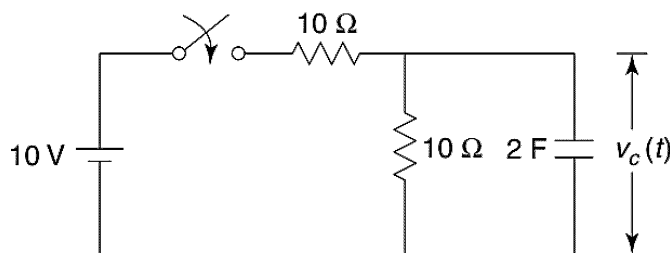
- B) Find the value of the current I_3 in the network shown **CO1, CO3** **6**



- C) What are initial conditions? Explain the initial conditions for Resistor, **CO1, CO3** **6**
capacitor and inductor

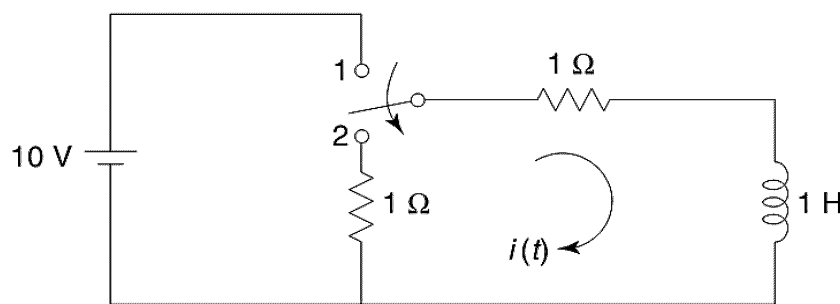
Q.3 Solve Any Two of the following.

- A) The switch in the network shown below switch is closed at $t = 0$. Determine **CO1** **6**
the voltage across Capacitor



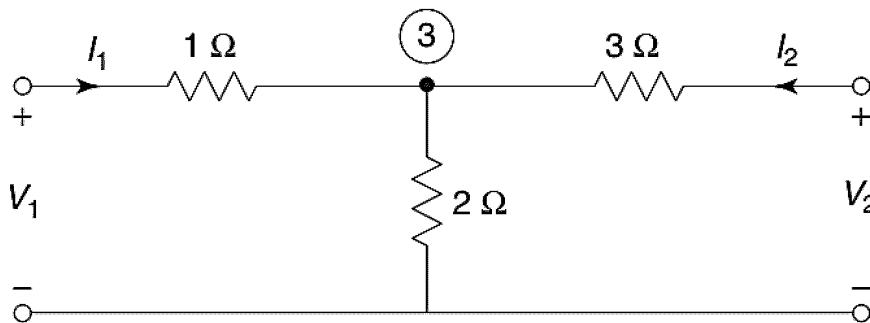
- B) Explain the Behaviour of basic elements in Laplace Transform **CO1** **6**

- C) In the network of Fig. given below , the switch is moved from the position 1 **CO1** **6**
to 2 at $t = 0$, steady-state condition having been established in the position 1.
Determine $i(t)$ for $t > 0$.



Q.4 Solve Any Two of the following.

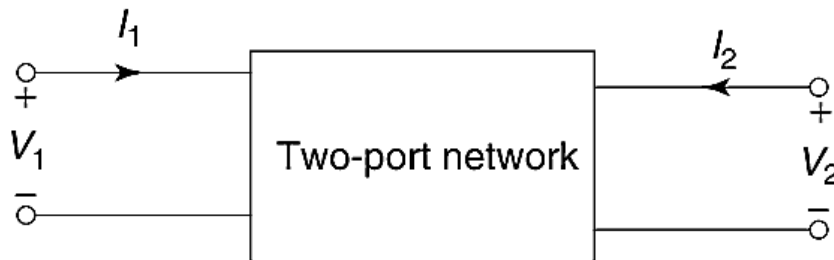
- A) Derive Y-parameter in terms of h-parameter and ABCD parameter **CO1, CO4 6**
- B) Find Y-parameters for the network shown. Determine whether the network is symmetrical and reciprocal **CO1, CO4 6**



- C) Define the terms: Transfer Impedance Function and Transfer Admittance Function. **CO1, CO4 6**

In the two-port network shown below, compute h-parameters from the following data:

- (a) With the output port short-circuited: $V_1=25V$, $I_1=1A$, $I_2=2A$
- (b) With the input port open-circuited: $V_1=10V$, $V_2=50V$, $I_2=2A$

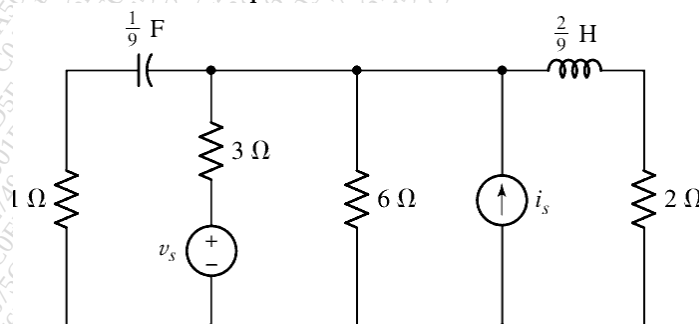


Q. 5 Solve Any Two of the following.

- A) Realise Cauer forms of the following LC impedance function. **CO1, CO3 6**

$$Z(s) = \frac{10S^4 + 12S^2 + 1}{2S^3 + 2S}$$

- B) Write a set of normal-form equations for the circuit **CO1, CO3 6**



- C) Explain about Propagation constant and Characteristic impedance in Π -network filters. **CO1, CO3 6**

***** End *****